MMC201 – Review of Question 7

One gradient-descent step for a one-hidden-layer network

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Sources. Problem statement: `11-nn-exercises.pdf`, Question 7. Hand derivation: `11-nn.rnote`, pp. 10–14 (source of the black content in this document).

Colour legend. Black: your derivation, transcribed from `11-nn.rnote` (pp. 10–14), and the problem statement. Red: my changes — shape corrections, the compact δ notation, the numerical values and the general rule. Notation $a \to b$ means "you wrote a, I replace it by b".

1 Problem

Statement taken from `11-nn-exercises.pdf`, Question 7.

$$Z_1 = W_1 X + B_1, \qquad H = \mathrm{ReLU}(Z_1), \qquad Y = W_2 H + B_2, \qquad E = \tfrac{1}{2}\|Y - Y_*\|^2,$$

$$X = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \; W_1 = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \; B_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \; W_2 = \begin{pmatrix} 2 & -1 \end{pmatrix}, \; B_2 = (0), \; Y_* = (1), \; \eta = 0.1.$$

Shapes: $X, B_1, Z_1, H \in \mathbb{R}^{2\times 1}$, $W_1 \in \mathbb{R}^{2\times 2}$, $W_2 \in \mathbb{R}^{1\times 2}$, $Y \in \mathbb{R}^{1\times 1}$.

Convention. Every gradient $\nabla_\theta E$ has the same shape as θ, so that $\theta_{\text{nuevo}} = \theta - \eta \nabla_\theta E$ is shape-compatible. $\mathrm{ReLU}'(0) = 0$ (as in Question 1).

2 Setup

$$\theta = \mathrm{FLAT}(W_1, B_1, W_2, B_2), \qquad \theta_{k+1} = \theta_k + \alpha_k p_k, \quad p_k = -\nabla_\theta E(\theta_k), \quad \alpha_k = \eta = 0.1.$$

Per parameter: $W_{\ell,\text{nuevo}} = W_\ell - \eta \nabla_{W_\ell} E$, $B_{\ell,\text{nuevo}} = B_\ell - \eta \nabla_{B_\ell} E$, $\ell = 1, 2$.

3 Derivation

3.1 $\nabla_{B_2} E$

Chain $B_2 \to Y \to E$:

$$\frac{\partial E}{\partial B_2} = \frac{\partial E}{\partial Y} \cdot \frac{\partial Y}{\partial B_2}, \qquad \frac{\partial E}{\partial Y} = Y - Y_* \in \mathbb{R}^{1\times 1} \; (1), \qquad \frac{\partial Y}{\partial B_2} = 1.$$

$$\frac{\partial E}{\partial B_2} = Y - Y_* \quad (2) \qquad =: \delta_2.$$

$\delta_2 = \nabla_Y E$ is the *output error signal*; it is reused in every other gradient.

3.2 $\nabla_{W_2} E$

Chain $W_2 \to Y \to E$:

$$\nabla_{W_2} E = \Big(\frac{\partial E}{\partial w_{2,1j}}\Big)_{j=1,2} \in \mathbb{R}^{2\times 1} \to \mathbb{R}^{1\times 2}, \qquad \frac{\partial Y}{\partial w_{2,1j}} = \frac{\partial}{\partial w_{2,1j}}\Big(\sum_{m=1}^{2} w_{2,1m} H_m + B_2\Big) = H_j,$$

$$\frac{\partial E}{\partial w_{2,1j}} = \frac{\partial E}{\partial Y} H_j, \qquad \nabla_{W_2} E = \frac{\partial E}{\partial Y} H \in \mathbb{R}^{2\times 1} \ (3) \to \ \delta_2 H^\top \in \mathbb{R}^{1\times 2}.$$

Reason: W_2 is 1×2, so its gradient must be a row. Entries unchanged.

3.3 $\nabla_{B_1} E$

Chain $B_1 \to Z_1 \to H \to Y \to E$:

$$\frac{\partial E}{\partial B_{1,j}} = \underbrace{\frac{\partial E}{\partial Y}}_{1\times 1} \underbrace{\frac{\partial Y}{\partial H}}_{1\times 2} \underbrace{\frac{\partial H}{\partial Z_1}}_{2\times 2} \underbrace{\frac{\partial Z_1}{\partial B_{1,j}}}_{2\times 1},$$

$$\frac{\partial Y}{\partial H} = W_2 \in \mathbb{R}^{1\times 2} \ (4), \qquad \frac{\partial H_i}{\partial z_{1,j}} = \delta_{ij}\, \mathbf{1}\{z_{1,j} > 0\} \ (5), \qquad \frac{\partial z_{1,i}}{\partial B_{1,j}} = \delta_{ij}.$$

$$\nabla_{B_1} E \to \frac{\partial E}{\partial B_1} = \begin{bmatrix} \frac{\partial E}{\partial B_{1,1}} \\ \frac{\partial E}{\partial B_{1,2}} \end{bmatrix}^\top = (Y - Y_*)\, W_2 \begin{bmatrix} \mathbf{1}\{z_{1,1} > 0\} & 0 \\ 0 & \mathbf{1}\{z_{1,2} > 0\} \end{bmatrix} \in \mathbb{R}^{1\times 2}$$

Let $r := \mathrm{ReLU}'(Z_1) = \begin{pmatrix} \mathbf{1}\{z_{1,1} > 0\} \\ \mathbf{1}\{z_{1,2} > 0\} \end{pmatrix}$, so that $\begin{bmatrix} \mathbf{1}\{z_{1,1} > 0\} & 0 \\ 0 & \mathbf{1}\{z_{1,2} > 0\} \end{bmatrix} = \mathrm{Diag}(r)$, and $\delta_2 := Y - Y_*$.

$$\begin{aligned}
\nabla_{B_1} E &:= \Big(\frac{\partial E}{\partial B_1}\Big)^\top \in \mathbb{R}^{2\times 1} \\
&= \Big[(Y - Y_*)\, W_2\, \mathrm{Diag}(r)\Big]^\top \\
&= \mathrm{Diag}(r)^\top W_2^\top (Y - Y_*)^\top && (ABC)^\top = C^\top B^\top A^\top \\
&= \mathrm{Diag}(r)\, W_2^\top\, \delta_2 && \mathrm{Diag}(r)^\top = \mathrm{Diag}(r), \ \ \delta_2 \in \mathbb{R}^{1\times 1} \Rightarrow \delta_2^\top = \delta_2 \\
&= \begin{bmatrix} r_1 & 0 \\ 0 & r_2 \end{bmatrix} \begin{bmatrix} (W_2^\top \delta_2)_1 \\ (W_2^\top \delta_2)_2 \end{bmatrix} \\
&= \begin{bmatrix} (W_2^\top \delta_2)_1\, r_1 \\ (W_2^\top \delta_2)_2\, r_2 \end{bmatrix} \\
&= (W_2^\top \delta_2) \odot r && (u \odot v)_i = u_i\, v_i \\
&= (W_2^\top \delta_2) \odot \mathrm{ReLU}'(Z_1) \ =: \ \delta_1 \in \mathbb{R}^{2\times 1}.
\end{aligned}$$

$$\frac{\partial Z_1}{\partial B_1} = I_2 \implies \nabla_{B_1} E = \nabla_{Z_1} E = \delta_1.$$

3.4 $\nabla_{W_1} E$

Chain $W_1 \to Z_1 \to H \to Y \to E$:

$$\nabla_{W_1} E = \Big(\frac{\partial E}{\partial w_{1,ij}}\Big)_{i,j=1,2} \in \mathbb{R}^{4\times 1} \to \mathbb{R}^{2\times 2}, \qquad \frac{\partial E}{\partial w_{1,ij}} = \underbrace{\frac{\partial E}{\partial Y}}_{1\times 2 \to 1\times 1} \underbrace{\frac{\partial Y}{\partial H}}_{1\times 2} \underbrace{\frac{\partial H}{\partial Z_1}}_{2\times 2} \underbrace{\frac{\partial Z_1}{\partial w_{1,ij}}}_{2\times 1},$$

$$\frac{\partial z_{1,k}}{\partial w_{1,ij}} = \frac{\partial}{\partial w_{1,ij}}\Big(\sum_{m=1}^{2} w_{1,km}x_m + B_{1,k}\Big) = \delta_{ki}\,x_j, \qquad \frac{\partial Z_1}{\partial w_{1,ij}} = \begin{bmatrix}\delta_{1i}\,x_j\\ \delta_{2i}\,x_j\end{bmatrix} \in \mathbb{R}^{2\times 1},$$

$$\frac{\partial E}{\partial w_{1,ij}} = (Y - Y_*)\,W_2 \begin{bmatrix}\mathbf{1}\{z_{1,1}>0\} & 0\\ 0 & \mathbf{1}\{z_{1,2}>0\}\end{bmatrix}\begin{bmatrix}\delta_{1i}\,x_j\\ \delta_{2i}\,x_j\end{bmatrix}$$

With $r = \mathrm{ReLU}'(Z_1)$, $\delta_2 = Y - Y_*$ and $\delta_1 = \mathrm{Diag}(r)\,W_2^\top \delta_2$ (Section 3.3):

$$\begin{aligned}
\frac{\partial E}{\partial w_{1,ij}} &= (Y - Y_*)\,W_2\,\mathrm{Diag}(r)\begin{bmatrix}\delta_{1i}\,x_j\\ \delta_{2i}\,x_j\end{bmatrix}\\
&= \big(\mathrm{Diag}(r)\,W_2^\top\,\delta_2\big)^\top\begin{bmatrix}\delta_{1i}\,x_j\\ \delta_{2i}\,x_j\end{bmatrix} && A^\top B^\top C^\top = (CBA)^\top,\ \mathrm{Diag}(r)^\top = \mathrm{Diag}(r),\ \delta_2^\top = \delta_2\\
&= \delta_1^\top\begin{bmatrix}\delta_{1i}\,x_j\\ \delta_{2i}\,x_j\end{bmatrix}\\
&= \begin{bmatrix}(\delta_1)_1 & (\delta_1)_2\end{bmatrix}\begin{bmatrix}\delta_{1i}\,x_j\\ \delta_{2i}\,x_j\end{bmatrix}\\
&= (\delta_1)_1\,\delta_{1i}\,x_j + (\delta_1)_2\,\delta_{2i}\,x_j\\
&= \sum_{k=1}^{2}(\delta_1)_k\,\delta_{ki}\,x_j\\
&= (\delta_1)_i\,x_j && \textstyle\sum_k a_k\,\delta_{ki} = a_i\\
&= \big(\delta_1 X^\top\big)_{ij} && (uv^\top)_{ij} = u_i\,v_j.
\end{aligned}$$

$$\nabla_{W_1}E = \begin{bmatrix}(Y - Y_*)\,W_2\begin{bmatrix}\mathbf{1}\{z_{1,1}>0\}\,x_1\\ 0\end{bmatrix}\\ (Y - Y_*)\,W_2\begin{bmatrix}\mathbf{1}\{z_{1,1}>0\}\,x_2\\ 0\end{bmatrix}\\ (Y - Y_*)\,W_2\begin{bmatrix}0\\ \mathbf{1}\{z_{1,2}>0\}\,x_1\end{bmatrix}\\ (Y - Y_*)\,W_2\begin{bmatrix}0\\ \mathbf{1}\{z_{1,2}>0\}\,x_2\end{bmatrix}\end{bmatrix} = \begin{bmatrix}(\delta_1)_1x_1\\ (\delta_1)_1x_2\\ (\delta_1)_2x_1\\ (\delta_1)_2x_2\end{bmatrix}$$

$$\begin{aligned}
\nabla_{W_1}E &= \begin{bmatrix}\dfrac{\partial E}{\partial w_{1,11}} & \dfrac{\partial E}{\partial w_{1,12}}\\ \dfrac{\partial E}{\partial w_{1,21}} & \dfrac{\partial E}{\partial w_{1,22}}\end{bmatrix} && \nabla_{W_1}E \in \mathbb{R}^{2\times 2}\\
&= \begin{bmatrix}(\delta_1)_1x_1 & (\delta_1)_1x_2\\ (\delta_1)_2x_1 & (\delta_1)_2x_2\end{bmatrix}\\
&= \begin{bmatrix}(\delta_1)_1\\ (\delta_1)_2\end{bmatrix}\begin{bmatrix}x_1 & x_2\end{bmatrix}\\
&= \delta_1 X^\top \in \mathbb{R}^{2\times 2}.
\end{aligned}$$

4 Summary in compact form

$$\delta_2 = Y - Y_*, \qquad \delta_1 = \big(W_2^\top\delta_2\big)\odot \mathrm{ReLU}'(Z_1),$$
$$\nabla_{W_2}E = \delta_2 H^\top, \quad \nabla_{B_2}E = \delta_2, \qquad \nabla_{W_1}E = \delta_1 X^\top, \quad \nabla_{B_1}E = \delta_1.$$

Gradient	Your notes	Compact form
$\nabla_{B_2} E$	$Y - Y_*$	δ_2
$\nabla_{W_2} E$	$\frac{\partial E}{\partial Y} H \in \mathbb{R}^{2\times 1}$	$\delta_2 H^\top \in \mathbb{R}^{1\times 2}$
$\nabla_{B_1} E$	$(Y - Y_*)\, W_2\, \mathrm{Diag}(\mathbf{1}\{Z_1 > 0\}) \in \mathbb{R}^{1\times 2}$	$\delta_1 \in \mathbb{R}^{2\times 1}$
$\nabla_{W_1} E$	four blocks, $\mathbb{R}^{4\times 1}$	$\delta_1 X^\top \in \mathbb{R}^{2\times 2}$

5 Numerical computation

Forward pass.

$$
\begin{aligned}
Z_1 &= W_1 X + B_1 = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \\
H &= \mathrm{ReLU}(Z_1) = \begin{pmatrix} \max(0,0) \\ \max(0,1) \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \\
Y &= W_2 H + B_2 = \begin{pmatrix} 2 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} + 0 = -1, \\
E &= \tfrac{1}{2}\|Y - Y_*\|^2 = \tfrac{1}{2}(-1-1)^2 = 2.
\end{aligned}
$$

Backward pass.

$$
\delta_2 = -2, \qquad W_2^\top \delta_2 = \begin{pmatrix} -4 \\ 2 \end{pmatrix}, \qquad \mathrm{ReLU}'(Z_1) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad \delta_1 = \begin{pmatrix} -4 \\ 2 \end{pmatrix} \odot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}.
$$

Caution: $z_{1,1} = 0$ lies exactly on the ReLU kink. With $\mathrm{ReLU}'(0) = 0$ the first hidden unit passes no gradient. Using $\mathrm{ReLU}'(0) = 1$ would give $\delta_1 = (-4, 2)^\top$ and different updates.

Gradients.

$$
\nabla_{B_2} E = -2, \quad \nabla_{W_2} E = \begin{pmatrix} 0 & -2 \end{pmatrix}, \quad \nabla_{B_1} E = \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \quad \nabla_{W_1} E = \begin{pmatrix} 0 \\ 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 2 & 4 \end{pmatrix}.
$$

Updates ($\eta = 0.1$).

$$
W_{1,\text{nuevo}} = \begin{pmatrix} 1 & -1 \\ -0.2 & 0.6 \end{pmatrix}, \quad B_{1,\text{nuevo}} = \begin{pmatrix} 1 \\ -1.2 \end{pmatrix}, \quad W_{2,\text{nuevo}} = \begin{pmatrix} 2 & -0.8 \end{pmatrix}, \quad B_{2,\text{nuevo}} = 0.2.
$$

Sanity check. With the new parameters, $Z_1 = (0, -0.2)^\top$, $H = (0, 0)^\top$, $Y = 0.2$, $E = \frac{1}{2}(0.8)^2 = 0.32 < 2$.

6 General rule

For $Z_\ell = W_\ell A_{\ell-1} + B_\ell$, $A_\ell = f_\ell(Z_\ell)$, $A_0 = X$, $\ell = 1, \ldots, L$:

$$
\delta_L = \nabla_{A_L} E \odot f_L'(Z_L), \qquad \delta_\ell = \left(W_{\ell+1}^\top \delta_{\ell+1}\right) \odot f_\ell'(Z_\ell), \qquad \nabla_{W_\ell} E = \delta_\ell A_{\ell-1}^\top, \qquad \nabla_{B_\ell} E = \delta_\ell.
$$

Question 7 is $L = 2$, $f_1 = \mathrm{ReLU}$, $f_2 = \mathrm{id}$, $A_1 = H$. Questions 4–6 are its single-layer pieces.